# CONTROL DE CALIDAD - SIKAGROUT 9400 BR | Blue Tech

## 1. Resumen Ejecutivo

Probetas analizadas: 138

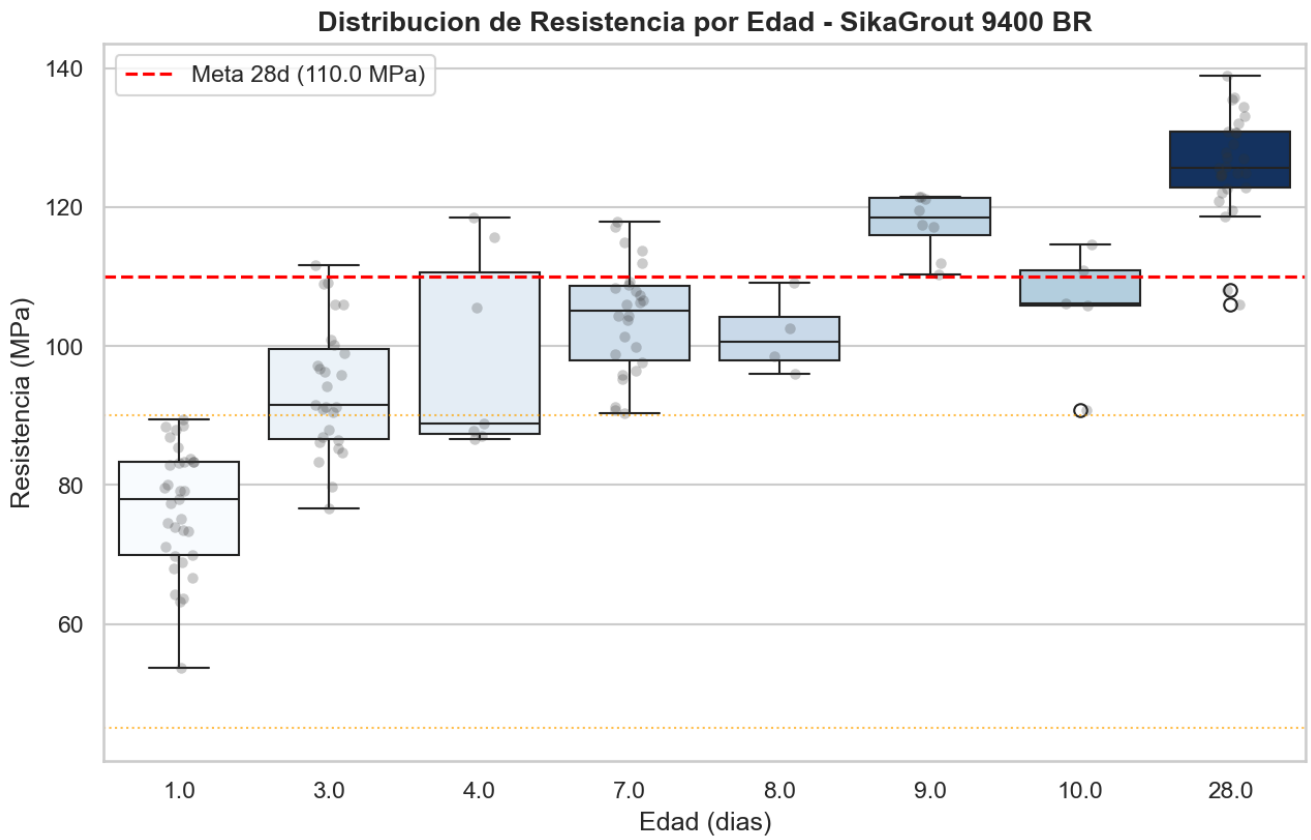
Resistencia Media (28d): 126.02 MPa

f'ck proyecto (Percentil 5): 113.84 MPa

f'ck requerido (Ficha): 108.00 MPa

Resultado final: APROBADO

| Edad (d) | N | Media | Desv. | Min | Max | CV% | Meta | Estado |
|---|---|---|---|---|---|---|---|---|
| 1 | 33 | 76.66 | 8.80 | 53.72 | 89.48 | 11.48 | 45 | PASA |
| 3 | 27 | 93.87 | 9.25 | 76.68 | 111.68 | 9.85 | - | --- |
| 4 | 7 | 98.57 | 14.28 | 86.64 | 118.48 | 14.49 | - | --- |
| 7 | 26 | 104.08 | 7.94 | 90.32 | 117.96 | 7.63 | 90 | PASA |
| 8 | 4 | 101.56 | 5.71 | 96.00 | 109.12 | 5.62 | - | --- |
| 9 | 8 | 117.58 | 4.36 | 110.28 | 121.52 | 3.71 | - | --- |
| 10 | 5 | 105.66 | 9.08 | 90.76 | 114.56 | 8.59 | - | --- |
| 28 | 28 | 126.02 | 7.40 | 106.00 | 138.84 | 5.88 | 110 | PASA |



## 2. Inferencia Estadistica (28 dias)

Prueba T vs 110.0 MPa: t = +11.4508, p = 7.1936e-12

Resultado T-test: SIGNIFICATIVO (p < 0.05)

f'ck calculado (Percentil 5): 113.84 MPa

f'ck requerido: 108.00 MPa

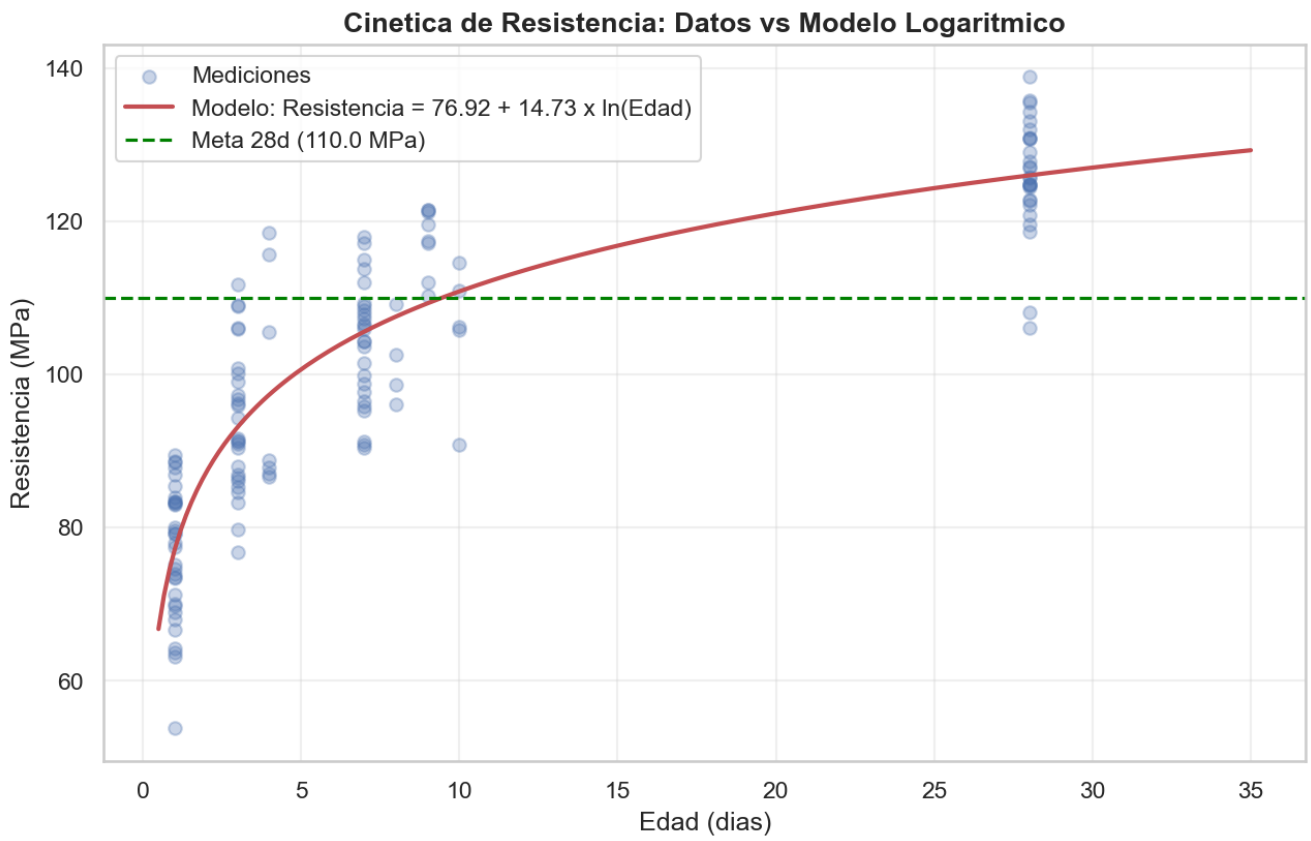
RESULTADO FINAL: APROBADO

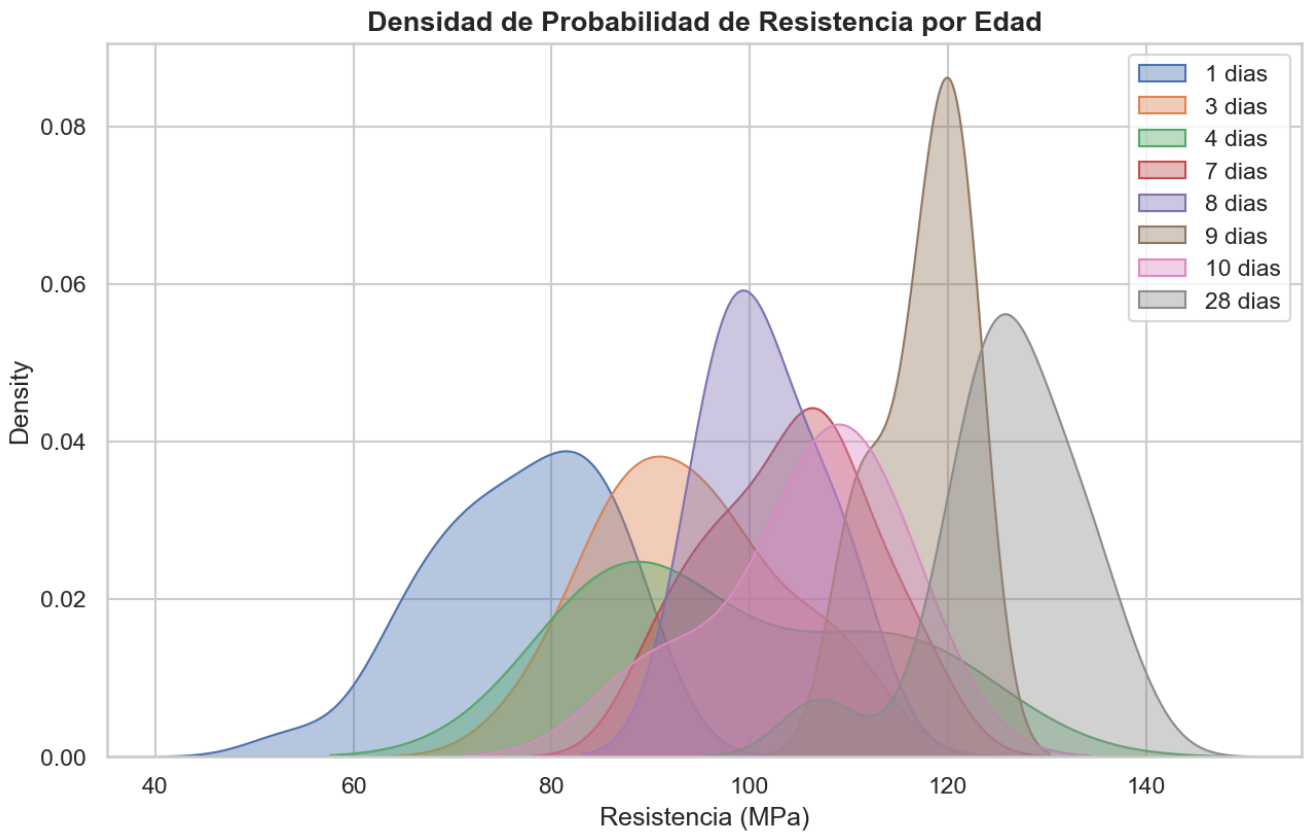
## 3. Modelo Predictivo Logaritmico

Ecuacion: Resistencia = 76.92 + 14.73 x ln(Edad)

R2 = 0.7931 | n = 138 probetas

Meta: >= 110.0 MPa a 28 dias

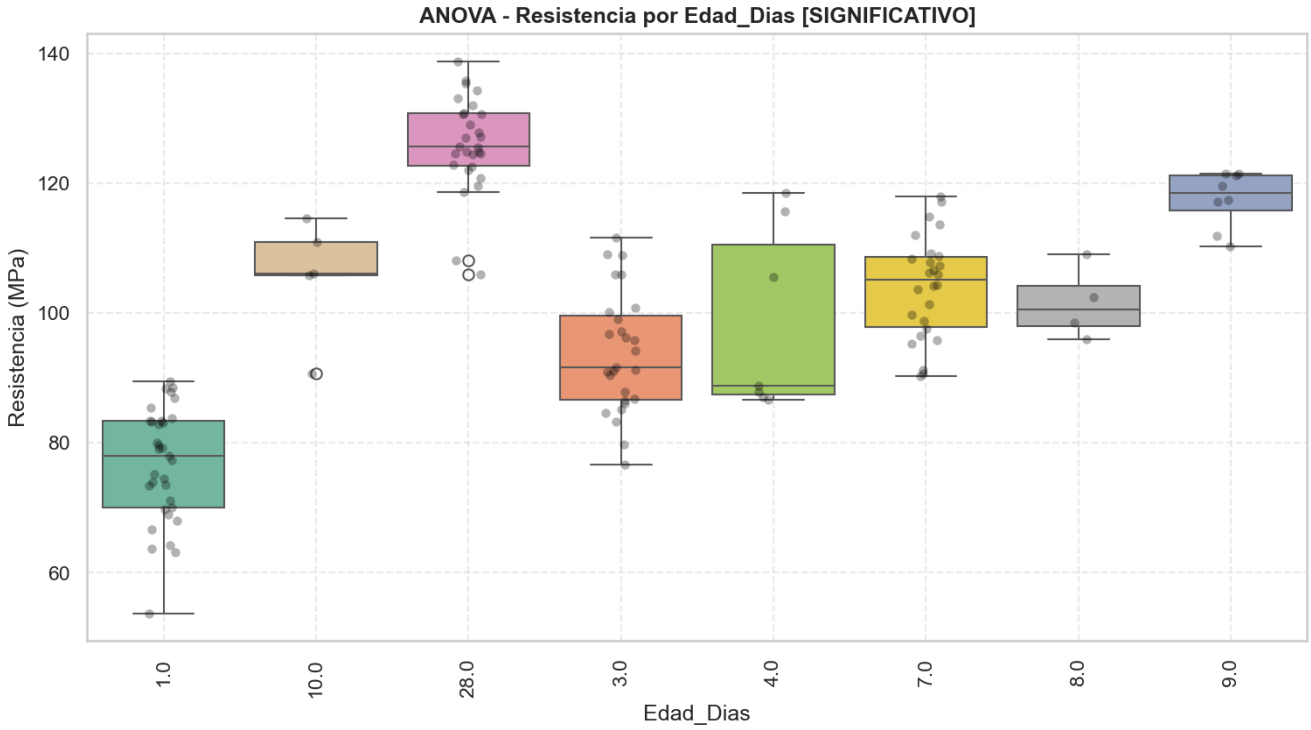



Densidad de Probabilidad de Resistencia por Edad
1 dias
3 dias
4 dias
7 dias
8 dias
9 dias
10 dias
28 dias
Density
0.00
0.02
0.04
0.06
0.08
40
60
80
100
120
140
Resistencia (MPa)

## 4. Analisis de Varianza (ANOVA)

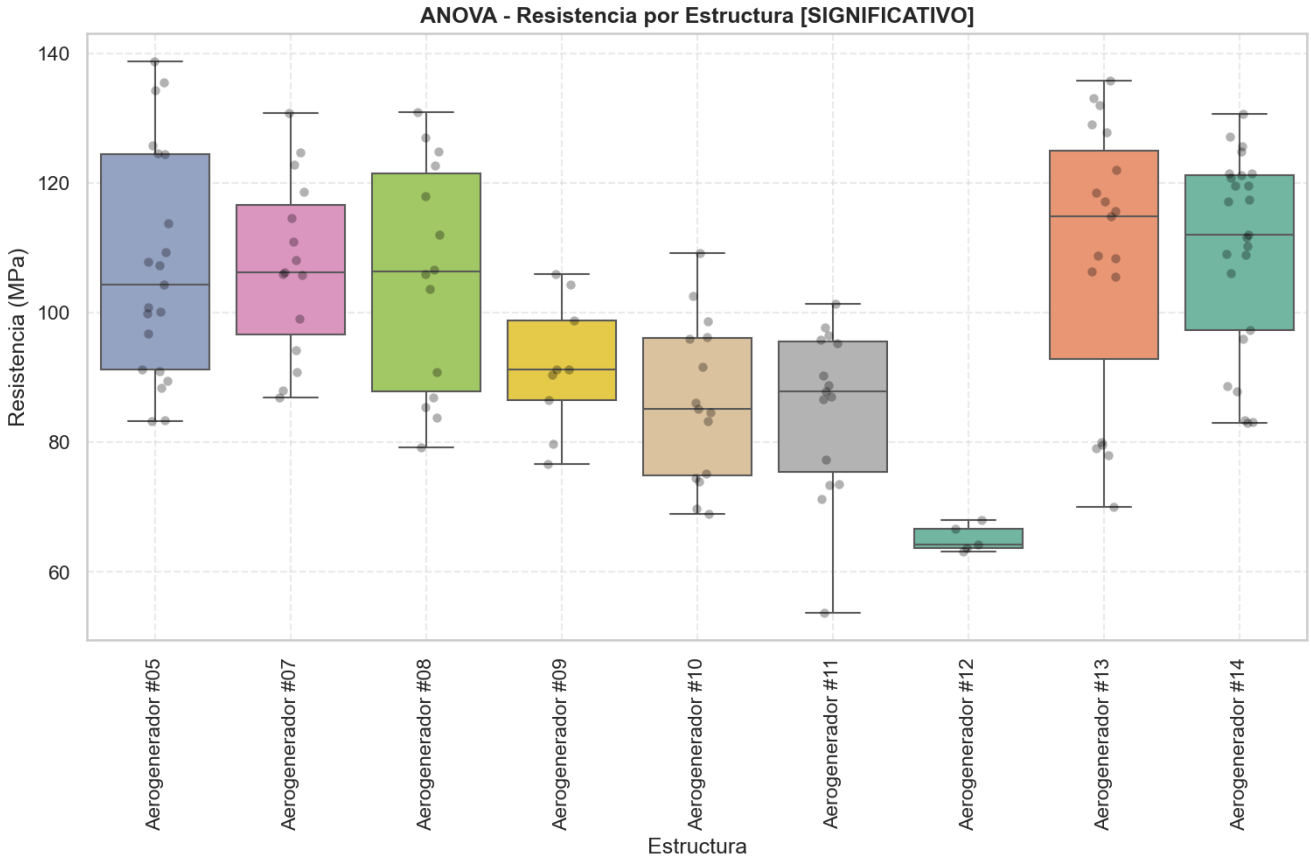
### Factor: Edad_Dias -> SIGNIFICATIVO

Shapiro-Wilk: p=0.9701 (Normal) | Levene: p=0.1790 (Homogeneo)




### Factor: Estructura -> SIGNIFICATIVO

Shapiro-Wilk: p=0.0622 (Normal) | Levene: p=0.0421 (No homogeneo (ADVERTENCIA))



## 5. Detalle del Reporte de Control de Calidad

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==================================================================================================
  REPORTE DE CONTROL ESTADISTICO - SIKAGROUT 9400 BR | Blue Tech
==================================================================================================

Probetas analizadas: 138
Fecha de generacion: 2026-02-26 11:06


==================================================================================================
  1. ESTADISTICA DESCRIPTIVA POR EDAD DE ROTURA
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  Edad     N      Media       Desv        Min        Max      CV%       Meta     Estado
--------------------------------------------------------------------------------------------------
     1    33      76.66       8.80      53.72      89.48    11.48         45       PASA
     3    27      93.87       9.25      76.68     111.68     9.85          -        ---
     4     7      98.57      14.28      86.64     118.48    14.49          -        ---
     7    26     104.08       7.94      90.32     117.96     7.63         90       PASA
     8     4     101.56       5.71      96.00     109.12     5.62          -        ---
     9     8     117.58       4.36     110.28     121.52     3.71          -        ---
    10     5     105.66       9.08      90.76     114.56     8.59          -        ---
    28    28     126.02       7.40     106.00     138.84     5.88        110       PASA

==================================================================================================
  2. INFERENCIA ESTADISTICA (28 DIAS)
==================================================================================================

  Prueba T vs 110.0 MPa:  t = +11.4508,  p = 7.1936e-12
  Resultado T-test: SIGNIFICATIVO (p < 0.05)

  Resistencia Caracteristica f'ck calculada (Percentil 5): 113.84 MPa
  Resistencia Caracteristica f'ck requerida (Ficha Tecnica): 108.00 MPa
  RESULTADO FINAL: APROBADO


==================================================================================================
  3. ANALISIS ANOVA ? Factor: Edad_Dias
==================================================================================================

                    sum_sq     df         F        PR(>F)
C(Edad_Dias)  41019.684171    7.0  79.82914  5.576211e-44
Residual       9542.807760  130.0       NaN           NaN

  Shapiro-Wilk (normalidad residuos):  p = 0.9701 -> Normal
  Levene (homogeneidad de varianzas):  p = 0.1790 -> Homogeneo

  Post-hoc Tukey HSD (alfa=0.05):
 Multiple Comparison of Means - Tukey HSD, FWER=0.05
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group1 group2 meandiff p-adj   lower    upper   reject
-----------------------------------------------------
   1.0   10.0   28.999    0.0  16.3314  41.6667   True
   1.0   28.0  49.3659    0.0  42.5836  56.1481   True
   1.0    3.0  17.2097    0.0  10.3598  24.0596   True
   1.0    4.0  21.9087    0.0  10.9245   32.893   True
   1.0    7.0   27.423    0.0  20.5011   34.345   True
   1.0    8.0   24.903    0.0  10.9278  38.8783   True
   1.0    9.0   40.928    0.0  30.5256  51.3305   True
  10.0   28.0  20.3669 0.0001   7.5513  33.1824   True
  10.0    3.0 -11.7893 0.0975 -24.6408   1.0622  False
  10.0    4.0  -7.0903 0.8496 -22.5465   8.3659  False
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  10.0    7.0   -1.576 0.9999 -14.4661  11.3141  False
  10.0    8.0   -4.096 0.9965 -21.8033  13.6113  False
  10.0    9.0   11.929 0.2303  -3.1193  26.9773  False
  28.0    3.0 -32.1562    0.0  -39.276 -25.0364   True
  28.0    4.0 -27.4571    0.0 -38.6117 -16.3026   True
  28.0    7.0 -21.9429    0.0  -29.132 -14.7537   True
  28.0    8.0 -24.4629    0.0 -38.5724 -10.3534   True
  28.0    9.0  -8.4379 0.2238   -19.02   2.1443  False
   3.0    4.0    4.699 0.8999  -6.4967  15.8948  False
   3.0    7.0  10.2133 0.0007   2.9604  17.4663   True
   3.0    8.0   7.6933 0.7026  -6.4488  21.8355  False
   3.0    9.0  23.7183    0.0  13.0928  34.3439   True
   4.0    7.0   5.5143    0.8  -5.7257  16.7543  False
   4.0    8.0   2.9943 0.9993 -13.5506  19.5391  False
   4.0    9.0  19.0193 0.0009   5.3578  32.6807   True
   7.0    8.0    -2.52 0.9994 -16.6972  11.6572  False
   7.0    9.0   13.505 0.0038   2.8328  24.1772   True
   8.0    9.0   16.025 0.0538  -0.1395  32.1895  False
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==============================================================================================
  4. ANALISIS ANOVA ? Factor: Estructura
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                     sum_sq     df        F        PR(>F)
C(Estructura)  18837.564418    8.0  9.57467  2.404868e-10
Residual       31724.927512  129.0      NaN           NaN

  Shapiro-Wilk (normalidad residuos):  p = 0.0622 -> Normal
  Levene (homogeneidad de varianzas):  p = 0.0421 -> No homogeneo (ADVERTENCIA)

  Post-hoc Tukey HSD (alfa=0.05):
            Multiple Comparison of Means - Tukey HSD, FWER=0.05
=========================================================================
      group1            group2      meandiff p-adj   lower    upper   reject
-------------------------------------------------------------------------
Aerogenerador #05 Aerogenerador #07  -0.0023    1.0 -16.7294  16.7249  False
Aerogenerador #05 Aerogenerador #08     -1.6    1.0 -18.6721  15.4721  False
Aerogenerador #05 Aerogenerador #09 -15.5054 0.2508 -35.2185   4.2077  False
Aerogenerador #05 Aerogenerador #10  -20.781 0.0044 -37.5081  -4.0538   True
Aerogenerador #05 Aerogenerador #11  -22.061 0.0018 -38.7881  -5.3338   True
Aerogenerador #05 Aerogenerador #12 -42.0263    0.0  -66.648 -17.4046   True
Aerogenerador #05 Aerogenerador #13   1.3678    1.0 -14.2986  17.0342  False
Aerogenerador #05 Aerogenerador #14   2.6169 0.9997 -12.0293  17.2631  False
Aerogenerador #07 Aerogenerador #08  -1.5977    1.0 -19.9849  16.7895  False
Aerogenerador #07 Aerogenerador #09 -15.5031 0.3238 -36.3655   5.3593  False
Aerogenerador #07 Aerogenerador #10 -20.7787 0.0119 -38.8461  -2.7113   True
Aerogenerador #07 Aerogenerador #11 -22.0587 0.0056 -40.1261  -3.9913   True
Aerogenerador #07 Aerogenerador #12  -42.024    0.0 -67.5751 -16.4729   True
Aerogenerador #07 Aerogenerador #13   1.3701    1.0 -15.7199  18.4601  False
Aerogenerador #07 Aerogenerador #14   2.6192 0.9999 -13.5408  18.7792  False
Aerogenerador #08 Aerogenerador #09 -13.9054 0.4946 -35.0454   7.2346  False
Aerogenerador #08 Aerogenerador #10  -19.181 0.0338 -37.5681  -0.7938   True
Aerogenerador #08 Aerogenerador #11  -20.461 0.0173 -38.8481  -2.0738   True
Aerogenerador #08 Aerogenerador #12 -40.4263 0.0001 -66.2046  -14.648   True
Aerogenerador #08 Aerogenerador #13   2.9678 0.9998   -14.46  20.3956  False
Aerogenerador #08 Aerogenerador #14   4.2169 0.9965 -12.2998  20.7337  False
Aerogenerador #09 Aerogenerador #10  -5.2756 0.9968  -26.138  15.5869  False
Aerogenerador #09 Aerogenerador #11  -6.5556 0.9861  -27.418  14.3069  False
Aerogenerador #09 Aerogenerador #12 -26.5209   0.07 -54.1193   1.0775  False
Aerogenerador #09 Aerogenerador #13  16.8732 0.1734  -3.1488  36.8952  False
Aerogenerador #09 Aerogenerador #14  18.1223 0.0817  -1.1119  37.3565  False
Aerogenerador #10 Aerogenerador #11    -1.28    1.0 -19.3474  16.7874  False
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Aerogenerador #10 Aerogenerador #12 -21.2453 0.1873 -46.7965   4.3058  False
Aerogenerador #10 Aerogenerador #13  22.1488 0.0024   5.0587  39.2388   True
Aerogenerador #10 Aerogenerador #14  23.3979 0.0004   7.2379  39.5578   True
Aerogenerador #11 Aerogenerador #12 -19.9653 0.2589 -45.5165   5.5858  False
Aerogenerador #11 Aerogenerador #13  23.4288  0.001   6.3387  40.5188   True
Aerogenerador #11 Aerogenerador #14  24.6779 0.0001   8.5179  40.8378   True
Aerogenerador #12 Aerogenerador #13  43.3941    0.0  18.5244  68.2638   True
Aerogenerador #12 Aerogenerador #14  44.6432    0.0  20.4033  68.8831   True
Aerogenerador #13 Aerogenerador #14   1.2491    1.0 -13.8102  16.3084  False
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  5. MODELO PREDICTIVO LOGARITMICO
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  Ecuacion: Resistencia = 76.92 + 14.73 x ln(Edad)
  R2 = 0.7931  |  n = 138 probetas
  Uso: MPa_28d = MPa_actual + 14.7251 x (ln(28) - ln(edad_actual))
  Meta a 28d: 110.0 MPa (Ficha Tecnica SikaGrout 9400 BR)
```